

Define Problem Statements

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	2 MARKS

1. Project Context

Evaluating national development is critical for identifying regional disparities, measuring policy outcomes, and planning international aid programs. The Human Development Index (HDI) is the global standard for this evaluation. However, the calculation process is traditionally slow, retrospective, and manual.

Stakeholders (such as government ministries, development agencies, and academic researchers) lack an accessible, real-time tool to simulate how changing key indicators (e.g., expanding primary school enrollment or increasing life expectancy through healthcare programs) affects a country's development rating.

2. Problem Statement Definition

2.1 The Core Problem

"Decision-makers in public policy and international development lack a predictive, interactive tool to simulate and evaluate the impact of national development indicators on overall Human Development Index (HDI) scores, leading to retroactive planning and sub-optimal resource allocation."

3. Specific Problem Components (P1 - P5)

Problem 1 (P1): Delayed and Retrospective Data

- **Description:** Official HDI rankings are published by the United Nations annually and represent historical data, often from the previous year or earlier. Policymakers cannot assess the immediate impact of current investments or policy changes in real time.
- **Impact:** Delayed policy responses and inability to react to current developmental trends.

Problem 2 (P2): Complexity in Manual Multi-dimensional Indexing

- **Description:** Calculating HDI manually requires normalization formulas for life expectancy, schooling indices, and log-transformed GNI values. This complexity presents a barrier for local government agencies and student analysts who lack specialized training.
- **Impact:** Dependency on external experts and slow reporting processes.

Problem 3 (P3): Inability to Perform "What-If" Scenario Analysis

- **Description:** Stakeholders cannot easily simulate hypothetical changes. For example, a policymaker cannot easily determine: *"If we raise the mean years of schooling from 6.2 to 8.5, what will be our developmental tier?"*
- **Impact:** Strategic planning is based on guesswork rather than data-driven simulations.

Problem 4 (P4): Lack of Actionable Regional Development Recommendations

- **Description:** Raw numerical indices do not tell policymakers *what to do*. Existing tools do not map predicted development scores to clear, actionable recommendations or developmental milestones.
- **Impact:** High analysis paralysis where indicators are identified but practical mitigation strategies are missing.

Problem 5 (P5): Poor Accessibility and Data Presentation

- **Description:** Existing statistical packages and data portals are complex, visual-heavy, or restricted behind paywalls, preventing local actors, student researchers, and public interest groups from accessing predictive tools.
- **Impact:** Limited democratic access to developmental planning tools.

4. Project Scope

4.1 In Scope

- **Regression Model:** Train and serialize a Linear Regression model using features (Country, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita) to predict continuous HDI values.
- **Automatic Classification:** Classify the predicted score into Very High, High, Medium, or Low tiers.
- **Insights Engine:** Map predicted tiers to qualitative policy suggestions and recommendations.
- **Responsive Interface:** Design a responsive web application for entering inputs and presenting results visually using gradient progress gauges.

4.2 Out of Scope

- **Real-time Government API Integration:** Automatically pulling live census or economic data from federal databases.
- **Macroeconomic Forecasting:** Simulating GDP growth or monetary inflation indices directly.

- **Sub-national and District-level Profiling:** The model runs on national indicators, not local municipality metrics.

5. Success Criteria

1. **Model Fit:** The regression model achieves an $R^2 \geq 0.85$ on test splits, indicating high predictive performance.
2. **Classification Accuracy:** The tier classification logic accurately maps continuous scores to standard UNDP categories.
3. **Response Latency:** Prediction calculations and result rendering complete in under 2 seconds.
4. **Usability Check:** User testing confirms that non-technical users can perform scenario inputs and understand recommendations.